

A Preliminary Investigation of Avatar Use in Video-Conferencing

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Figure 1: Video call between participant (left) and experimenter (right), both embodied in virtual avatars.

ABSTRACT

Avatar use on video-conference platforms has found dual purpose in recent times as a potential method for ensuring privacy and improving subjective engagement with remote meeting, provided one can also ensure a minimal loss in the quality of social interaction and sense of personal presence. This preliminary study focuses on interaction through virtual avatars in a video conferencing context.

Index Terms: Human-centered computing—Virtual reality—;—Computing methodologies—Perception—; Computing methodologies—Motion capture—

1 INTRODUCTION

The interaction between humans and avatars in virtual environments has been shown to produce cognitive reactions which can be scaled in comparison with real world social interactions. The work outlined in the present study sought to measure previously defined effects that are known to occur in virtual interactions. Our novel approach to perception and recognition of self-face avatars is especially relevant in the current era of remote working. Much of the contemporary research in avatar interaction focuses on the concept of Embodiment, particularly on the influence of control, appearance and the point of view dimensions, which are considered to be the three factors which govern the ‘Sense of Embodiment’ (SoE) [1]. Another prominent feature of this field is avatar-induced behavioural change, known as the ‘Proteus Effect’ [4]. In designing this study, we sought to quantify and measure the SoE, Self-face recognition and perceived Social Presence. We also predicted there would be evidence for a positive character affect, which could be indicative of a detection of the Proteus effect.

2 EXPERIMENT

The metrics used for this design have been derived from a body of similar research, using methods adapted from established evaluations of social presence, enfacement and embodiment in avatar interaction settings. The experiment was conducted within participants with the same male experimenter for all of the recorded versions. Fourteen participants were asked to engage in a video-conversation using

a self-representative avatar for the entire experiment. The conversations took place in two separate periods following two separate Mood Induction Procedures (MIP). The participants were individually welcomed into a pre-arranged conference call, where they used a modern avatar application with accurate tracking of 198 facial landmarks and moderately rich facial animation of 50 expression blendshapes. The conversations for each participant were guided using two different videos, chosen to elicit either a positive or negative emotional response. After each of the two videos, the participant and experimenter engaged in four minutes of conversation about the videos, in order to encourage the participants to fully engage in the social use of their avatar (See Figure 1). After conversation, participants answered questions adapted from previous studies [2,3] and provided data via a comment section.

3 RESULTS

The participant questionnaire data was collected via a 7-Point Likert-Scale, so a the analysis for these results was conducted using a one sample t-test of means against the constant of 4 (see Figure 2). This statistical test helped us to elucidate the noisy data that was collected, which consisted of a single questionnaire and comment section for each participant. While a t-test is valuable for determining causal efficacy, it also served to focus our data collection towards our outlined hypotheses. A between-groups ANOVA was conducted to determine the effect of quality of facial tracking (6 Low v 8 High) which was an unavoidable factor in these remote experiments.

Our questionnaire results provided evidence for an interaction between the valence stimuli and the moods of participants, as demonstrated by our measurements for Affect Level ($M = 4.57, SD = 1.82$). The impact of positive valence was more prevalent in the results than the effect of negative valence. This is evidenced by the measurements taken for the Positive Expression ($M = 5.43, SD = 1.28, t(14) = 4.16, p = 0.001$). However, there was no significant effect detected for the Perceived Positive Affect measure ($M = 4.21, SD = 1.76, t(14) = 0.45, p = 0.65$). There is a contrast here with the results for the other question items on negative valence, with the strongest influence being observed in the Negative Expression question data ($M = 3.07, SD = 1.35$), meaning that it was not difficult to express emotions after the negative video ($t(14) = -2.32, p = 0.036$). The other values for the negative valence questions provide even lower indications of negative affect.

3.1 Ownership

The results for face and body ownership questions between subjects indicated that senses of enfacement and embodiment were not felt

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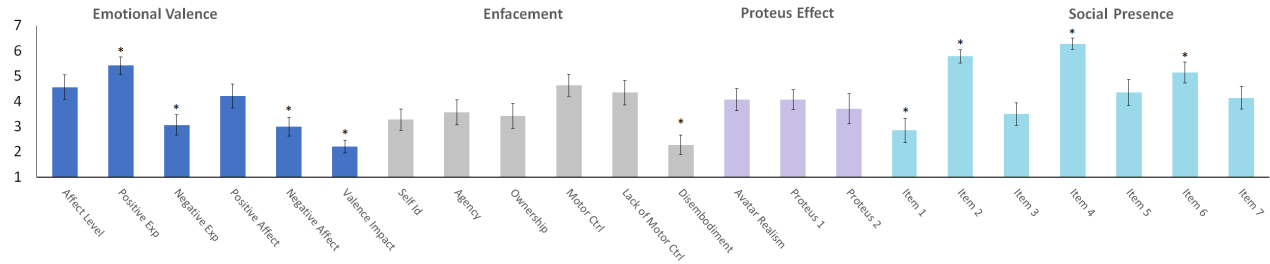


Figure 2: Results for questions Emotional Valence, Enfacement, Proteus Effect and Social Presence. Error bars show the standard error of the means. X-Axis labels are the variable names for the measured items. Star indicates statistical significance < 0.05 against the reference value '4'.

by the participants. The highest means in the data gathered on SoE come from the Motor Control ($M = 4.64, SD = 1.65$) and Lack of Motor Control ($M = 4.36, SD = 1.82$) measures. These results are consistent with previous studies on the topic, and confirm the importance of movement for SoE. Face-ownership effects were not observable from the data on our ownership items ($M = 3.43, SD = 1.76$), and the face-control measure that intended to augment the data for enfacement ($M = 3.57, SD = 1.80$).

3.2 Proteus Effect

We conducted a non-parametric Spearman's Rank Test to examine correlations between the results on enhanced feelings of attractiveness and ability to talk easily, friendliness, awkwardness, and talkative ratings, as a means to identify evidence for the Proteus Effect. There were no findings in the Spearman's results to indicate a main effect. Interestingly, previous studies that attempt to quantify this effect in computer-mediated dyadic avatar interaction have also failed to demonstrate this effect, and so our evidence seems indicative of an upper boundary or a necessary condition that is missed in-between immersive VR and dyadic remote conversation.

T-test results indicate that there was no main effect for the Proteus 2 item ($M = 3.71, SD = 2.23$) from the reference value of 4 ($t(14) = -0.47, p = 0.64$) indicating the participants did not feel significantly more talkative with their avatar. There was a similarly unconvincing result for the Proteus 2 item on attractiveness ($M = 4.07, SD = 1.49$) that reflected no major effect ($t(14) = 0.18, p = 0.86$). These results are in line with the information derived from the Spearman's correlation results.

3.3 Presence

The questions where social presence was most visible were for other-presence items for Item 2 'Stimulation' ($M = 5.79, SD = 0.94$) and Item 4 'Attention' ($M = 6.29, SD = 0.79$). The t-test results for these items confirmed high levels of statistical significance, with Item 4 ($t(14) = 10.36, p = 0.00$) slightly stronger than Item 2 ($t(14) = 6.85, p = 0.000012$). Social Presence, by these measures, can be confirmed as an observed result. The participants also evaluated trust in the other avatar based on Item 6 t-test results ($M = 5.14, SD = 1.56, t(14) = 2.73, p = 0.017$). This is consistent with the appearance of these avatars, as larger eyes and face width have been associated with heightened senses of trust. Conversely, the participants did not perceive empathy in the avatar face ($M = 4.14, SD = 1.65, t(14) = 0.32, p = 0.75$).

3.4 Quality

A between-groups ANOVA was conducted to determine the effect of quality of facial tracking and we found a main effect of quality on attractiveness ratings ($F(1, 12) = 5.36, p < 0.04$), where participants felt more attractive as their avatar when the quality was high than when it was low. This result is consistent with our finding that face

animation quality did not impact agency or ownership under these conditions, although our study did not deliberately manipulate this factor. Another main effect of quality was found where participants found it more awkward using the avatar after the negative video when the facial tracking was poor ($F(1, 12) = 5.8, p < 0.04$).

4 CONCLUSIONS

Positive affect scores suggest that the use of avatars for post-positive conversation was of higher quality, and that participants were more likely to feel friendly. This result is consistent with similar studies, which provide strong evidence for the influence of positively affected avatars. The results also indicate that positive valence was not impacted by a reduced quality of facial animation accuracy. Conversely it seems that there was an impact on negatively valenced questions. This effect is consistent with previous related studies, which have shown how accurate real-time facial mapping prompts higher levels of reported appeal towards a virtual face. The effect of self-face recognition, ownership and control were not observed, at least at the level we had predicted based on other studies on ownership and enfacement. Scores were comparable for our Sense of Embodiment and Proteus Effect data, although neither phenomenon was observed above chance, which could implicate SoE as a necessary precondition for the effect of transformed self-representation on behaviour. The use of these avatars was not measured to detract from the social presence you would expect in a regular face-to-face meeting, and scores for this item were high in general. Recording behavioural measures of social presence, such as mimicry of physical movements, through analysis of animation data and movement data correlations could be a useful point of future convergence for this kind of work. We believe subsequent research should seek to consolidate our results with investigations into visual effects on valence outcomes. Previous work in the area has suggested that positive and negative affect conditions can be enhanced by bright and dark key-light illumination respectively. Self-face recognition, mood induction processes are also areas for which our evidence presents grounds for future experimentation.

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